

7-2 Types of Tests

The project applies the testing types required by the graduation-project report guide, with automated backend focus and manual acceptance on the UI.

Test Type	Application in UPPMS	Tool / Approach
Unit Testing	FitnessCalculator, EvolutionaryOperators, TrackService, XmlImportService, AI services	PHPUnit Unit (50 cases)
Integration Testing	Services with database and Eloquent models	Feature/Unit with DatabaseTransactions
System Testing	Full API flows: proposals, tracks, committees, XML, scheduling	PHPUnit Feature (91 cases)
Acceptance Testing	Admin / Supervisor / Student scenarios on SPU campus	Manual runs + SpuCampusDemoSeeder data
Performance / Stress	Genetic scheduling engine on dozens of projects	GA runs on Campus Demo dataset
Security Testing	Tenant isolation, RBAC, XML matching, pending-user blocking	Feature tests + Middleware review
Usability Testing	Clarity of proposals, track timeline, scheduling, XML import UI	User testing + notes
Compatibility Testing	Modern browsers (Chrome/Edge) and RTL/LTR modes	Manual UI verification
Regression Testing	Re-run suite after major fixes	php vendor/bin/phpunit

Total automated test cases in the repository: **141** (Unit: 50 — Feature: 91).

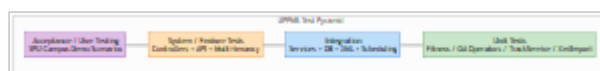


Figure 7-2: UPPMS Test Pyramid

Image file: diagrams-en/02-test-pyramid.png